



MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING & INFORMATION TECHNOLOGY

UniGuide – A UNIVERSITIES REVIEWS APPLICATION

BY

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CIT-227/032/2021

MAY, 2025

Submitted in partial fulfillment of the requirements of Fourth Year Bachelor of
Science in Software Engineering/Computer Science.

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DECLARATION

I hereby declare that this Project is my own work and has, to the best of my knowledge, not been submitted to any other institution of higher learning.

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This project has been submitted as a partial fulfillment of requirements for the Bachelor of Science in Software Engineering of Multimedia University of Kenya with my approval as the University supervisor.

Supervisor: Dr. Peter Muturi

Signature:

Date:

ACKNOWLEDGMENT

I acknowledge the following who have been instrumental throughout this project proposal:

- The school, Multimedia University of Kenya for being my learning institution.
- I also acknowledge my personal efforts and commitment throughout this project.
- My supervisor, Dr. Peter Muturi for being a great guide and enabling me acquire a different insight and approach to software engineering through this project.

ABSTRACT

This project introduces **UniGuide**, a comprehensive web-based application developed using Java (Spring Boot), MySQL, and Thymeleaf. UniGuide is designed to address the need for a centralized platform where prospective and current university students in Kenya can access and share authentic reviews about universities. The system enables users to post reviews, interact with existing ones and view fun, AI-generated facts for each institution. Each university profile includes key details such as a brief overview, courses/programs offered and quick AI insights to enhance user accessibility

The motivation for this project stems from the challenges students face in accessing honest, peer-generated insights about universities—information often overlooked in official marketing materials. UniGuide bridges this gap by facilitating user engagement and promoting informed decision-making to foster transparency in higher education.

This project demonstrates the practical integration of backend development, database management, user authentication, and AI-powered features in solving real-world problems. It also contributes to the broader goal of improving the Kenyan higher education experience through technology.

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List of Abbreviations

Table 1: List of abbreviations

Abbreviation	Full form
AI	Artificial intelligence
API	Application programming interface
AWS	Amazon web services
CIT	Computing & information technology
CSS	Cascading style sheets
CSRF	Cross-site request forgery
ERD	Entity relationship diagram
IDE	Integrated development environment
JPA	Java persistence API
KEI	Knowledge Exchange Institute
KCSE	Kenya Certificate of Secondary Education
MVC	Model-View-Controller
MySQL	My Structured Query Language
PWA	Progressive Web App
RAD	Rapid Application Development
SMTP	Simple Mail Transfer Protocol
UI/UX	User Interface/User Experience

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Chapter 1 – Introduction

1.1 Background of Study

Selecting the right university is a significant challenge for students in Kenya, particularly those from remote areas with limited access to reliable information. Many students rely on word-of-mouth, outdated websites, and rankings that fail to provide firsthand insights into universities. This lack of proper guidance often leads to uninformed decisions, resulting in consequences such as course changes, lack of motivation, and, in extreme cases, dropouts. According to **Zajac, T., Perales, F., Tomaszewski, W., Xiang, N., & Zubrick, S. R. (2023). Student mental health and dropout from higher education: an analysis of Australian administrative data. *Higher Education*, 87(2), 325–343. <https://doi.org/10.1007/s10734-023-01009-9>** , approximately 15% of students drop out of university within their first academic year.

These issues often stem from a lack of purpose or motivation within the educational ecosystem. With the increasing reliance on digital platforms for educational decision-making, there is a growing need for a localized, user-friendly solution tailored specifically to Kenyan students. This project aims to address this gap by providing a comprehensive platform where students can access real reviews and ratings from peers, current students, and alumni, empowering them to make informed and wise choices about their higher education.

1.2 Problem Statement

Students in Kenya face significant challenges in selecting the right university due to the lack of reliable, firsthand information. Existing platforms lack a dedicated focus on university education, leaving students to depend on insufficient resources such as word-of-mouth, outdated websites, and generalized rankings. This often leads to uninformed decisions, resulting in dissatisfaction, disappointment, and severe consequences such as financial losses from course changes. These outcomes not only affect individual students but also have broader economic and social implications, including wasted resources and a less skilled workforce. This project seeks to address these challenges by providing a localized, user-friendly platform where students can access real reviews

and ratings from peers, current students, and alumni, empowering them to make informed and purposeful decisions about their higher education.

1.3 Aim of the Study

To develop a web-based platform that provides students with reliable reviews, ratings, and detailed information about universities in Kenya.

1.3.1 Research Objectives

- To create a platform where students can post and access reviews on universities based on various criteria such as location, programs/courses offered, facilities and amenities, campus life quality, student support services and career prospects.
- To provide detailed institution profiles, including history, programs and facilities.
- To include fun facts about universities using AI-generated insights thus enabling potential students gain new exciting insights about the particular institution.
- To offer user accounts for posting and interacting with reviews.
- To evaluate the platform's effectiveness in aiding students' decision-making process in choosing the correct universities and career paths.

1.4 Significance/Justification of the Study

This study is significant as it addresses the critical information gap faced by students, enabling them to make better-informed academic decisions. By offering transparent and honest reviews, the platform could encourage universities to improve their services, ultimately raising the overall quality of universities education in Kenya. Additionally, the platform has the potential to scale into a Software-as-a-Service (SaaS) model, with strategies such as advertising, premium features for universities, and sponsored content.

1.5 Scope

The system will cover the major universities in Kenya. The platform is designed exclusively for Kenyan students and universities, with no immediate plans for expansion into other markets. This localized approach ensures that the platform remains tailored to the needs of its target audience.

1.6 Assumptions

- Users will provide honest and constructive feedback/reviews.
- Universities will be interested in verifying and enhancing their profiles.
- Students will actively engage with the platform to make informed decisions.

1.7 Limitations

- Potential bias in user-generated content.
- Internet access limitations in remote areas.
- Resistance from universities wary of negative reviews.
- The application will only be accessible as online content in data heavy devices such as smartphones, laptops, and computers.
- The application will not be downloadable and installable in a user's device.

Chapter 2 – Literature Review

2.1 Introduction

This chapter reviews existing platforms that provide user-generated reviews and their applicability to universities education. A thorough examination of related systems provides context for the current problem, highlighting their strengths and weaknesses.

2.2 Related Systems

1. **The Knowledge Exchange Institute (KEI)-** is an organization that offers study-abroad programs, including opportunities in Kenya. They collaborate with universities like the United States International University (USIU) in Nairobi, offering courses in various academic disciplines. KEI provides cultural experiences, internships, and volunteer opportunities, supporting students throughout their academic journey.
2. **UniversityGuru-** is an online platform that provides rankings, ratings, and insights into universities in Kenya and other countries. It evaluates universities based on web metrics, student opinions, and academic performance indicators, helping prospective students make informed decisions about their education.
3. **EDUopinions-** is a worldwide review platform that allows students from different countries to share their experiences and opinions about universities and colleges. It provides insights into various aspects such as academic quality, campus facilities, and student life, helping prospective students make informed decisions.

2.3 Limitations/Weaknesses of These Systems

While the above applications provide valuable information, they have several weaknesses compared to my proposed application:

1. International Focus with Limited Kenyan-Specific Data

These platforms primarily focus on global education, offering only minimal details about Kenyan universities. This lack of focus makes it difficult for students to find detailed and relevant institution-specific information tailored to the Kenyan context.

2. Limited Scope and Features

UniversityGuru primarily provides rankings based on web metrics and lacks comprehensive insights into student experiences, campus life, and facilities. Additionally, it does not include all Kenyan universities, making it less comprehensive.

3. Tedious Registration Processes

Platforms like EDUopinions have a lengthy and cumbersome sign-in process, which can discourage users from sharing their experiences and accessing information quickly.

4. Limited Regional Customization

These platforms do not cater to the specific needs of Kenyan students, such as cultural aspects, funding opportunities, and course-specific details that are crucial when selecting universities.

2.4 How the Proposed Solution Will Handle These Weaknesses

1. Kenyan-Specific Focus

This application will provide detailed and relevant information tailored exclusively to Kenyan universities, ensuring comprehensive coverage with institution-specific insights.

2. Comprehensive Institution Profiles

Unlike platforms that focus mainly on rankings, the app will offer in-depth details on universities, including programs, facilities, campus life, and student support services, ensuring users have all the information they need in one place.

3. User-Friendly Access

The app will feature a simplified and quick sign-in process, minimizing barriers to access and encouraging more users to share and interact with reviews effortlessly.

References

Compare universities & rankings. (n.d.). Universityguru. Retrieved January 27, 2025, from <https://www.universityguru.com/>

Education Opinions Advisor SL. (2024, September 26). *Student Reviews & University opinions. Business School Rankings | EDUOpinions.*

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Kenya – KEI Abroad. (n.d.). <https://www.keiabroad.org/kenya/>

Chapter 3 – Methodology

3.1 Introduction

This chapter outlines the methodology that will be used to develop the platform, ensuring the project meets its objectives effectively.

3.2 Methodology

The Rapid Application Development (RAD) methodology will be employed for the system's development. RAD emphasizes iterative development and user feedback, making it suitable for building user-focused applications like this platform.

Implementation of RAD Methodology

The implementation of RAD for this project will involve the following steps:

1. **Requirements Planning:** Initial requirements will be gathered through surveys, online research, and personal insights. These requirements will inform the initial design and functionality of the system.
2. **User Design (Prototyping):** A working prototype of the platform will be developed using Java Spring Boot for the backend, MySQL for the database, and Thymeleaf with Tailwind CSS for the frontend. Users will be involved in evaluating the prototype and providing feedback.
3. **Iterative Development:** Based on user feedback, the prototype will be refined through multiple iterations. Each iteration will focus on enhancing features such as review submission, interaction options (likes/dislikes), and detailed institution profiles.
4. **Finalization and Deployment:** After thorough testing and user validation, the system will be finalized and deployed for public use. Training materials and documentation will also be provided to ensure ease of use.

Justification

RAD is ideal for this project because it:

- Accommodates changes in user requirements during development.

- Ensures the platform aligns with the needs and expectations of the target audience.
- Speeds up the development process compared to traditional methodologies.

Justification for AI-Generated Fun Facts Feature:

The inclusion of AI-generated fun facts in UniGuide aims to enhance the platform's engagement by offering a more dynamic and interactive experience. Rather than providing only traditional, factual data, the fun facts serve to make the university selection process more interesting and relatable. This feature is designed to add a personal touch, helping students feel more connected to the institutions they are exploring and making the platform more enjoyable overall.

3.3 Data Collection Methods and Data Sets

To ensure the system meets its objectives, data will be collected using the following methods:

1. **Surveys:** Surveys will be conducted with students and alumni to gather insights about their experiences and expectations. For example, students in the **CIT** faculty at Multimedia University of Kenya have been engaged, and they have approved the project. Additionally, recent high school graduates (Form Four leavers) from the **2024 KCSE** cohort will be surveyed to understand their needs and preferences.
2. **Online Research:** Data will be collected from institutional websites, educational bodies, and existing platforms to provide accurate and comprehensive institution profiles.
3. **Interviews and Focus Groups:** Interviews with current students, alumni, and academic advisors will provide deeper insights into the challenges students face when selecting universities.

3.4 Moderation and Verification System

To ensure the quality and authenticity of reviews, the platform will implement a robust moderation and verification system. Users will be categorized into two types during sign-up:

- **Students:** These are students currently pursuing their studies in that particular institution.
- **Alumni:** Past students who have already graduated but would like to offer some insights on their alma mater or just discuss their time there.

Connection to Project Objectives

- Surveys will inform the development of user-focused features such as review submission, fun fact displays, and interaction options.
- Online research will ensure accurate and comprehensive institution profiles, including programs, facilities, and locations.
- Interviews and focus groups will help refine the platform's design by incorporating real user needs and expectations, ensuring the system addresses key challenges in the university selection process.

3.4 Tools and Technologies

- **Backend:** Java Spring Boot for efficient and scalable backend development.
- **Database:** MySQL for managing data related to universities, reviews, and user interactions.
- **Frontend:** Thymeleaf with Tailwind CSS for creating dynamic and responsive user interfaces.
- **Integrated Development Environment (IDE):** IntelliJ IDEA for coding and debugging.

- **Version Control:** Git and GitHub for tracking changes and collaborating on code development.
- **AI-Powered Fun Facts:** AI-driven tool that generates unique facts about universities, using Groq's LLM free API **llama-3.3-70b-versatile** model. This interactive feature enhances user engagement by providing interesting and relevant insights on each institution's profile page.

Chapter 4-System Analysis

This chapter presents a detailed analysis of the UniGuide system, including current system state, flowcharts, system requirements, and various diagrams that illustrate the system's architecture and functionality.

4.1 Detailed Analysis of Current System

4.1.1 System Overview

UniGuide is a web-based university review platform that allows users to browse university information, generate AI-powered fun facts, and post reviews with proper authentication and authorization mechanisms. The system follows a layered architecture with clear separation between presentation, business logic, and data layers.

4.1.2 Flow Chart

The following flowchart illustrates the primary user journey through the UniGuide system:

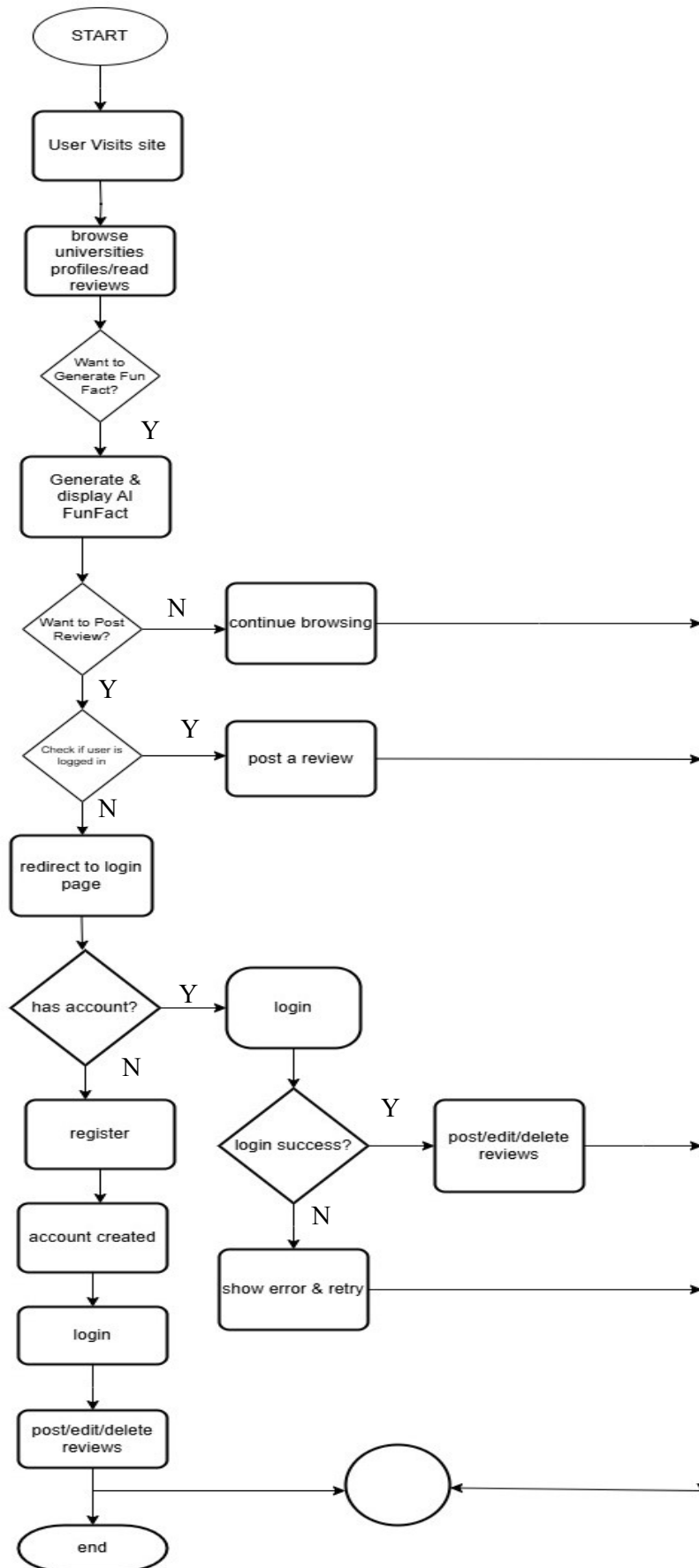


Figure 1:flowchart

4.2 System Requirements

4.2.1 Functional Requirements

4.2.1.1 User Management

- The system shall allow users to register with their name, valid student email, password, university name and user type (Student or Alumni).
- The system shall authenticate users using student email and password.
- The system shall maintain user sessions for logged-in users.

4.2.1.2 University Information Management

The system shall display comprehensive university profiles including:

- University name and description
- Programs/courses offered (auto-updated via web scraping)
- Reviews posted by users
- Gen AI facts for users to gain a quick insight of a university

4.2.1.3 Data Scraping and Management

- The system shall automatically scrape course information from university websites
- The system shall schedule regular scraping tasks to keep course data updated
- The system shall parse and render course details including:
 - Course names
 - Department/faculty information
- The system shall handle scraping failures gracefully with fallback mechanisms

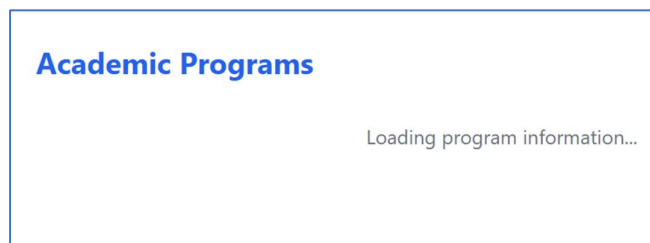


Figure 2:A screenshot showing an example of a fallback mechanism of the web scrapping service

4.2.1.4 Review Management

- The system shall allow authenticated users to post reviews for universities
- The system shall allow users to edit their own reviews
- The system shall allow users to delete their own reviews
- The system shall display reviews with timestamps
- The system shall implement review rating system (1-5 stars)
- The system shall prevent users from deleting or editing other users' reviews

4.2.1.5 AI-Powered Features

- The system shall generate fun facts about universities using AI (Groq API)
- The system shall handle AI service failures gracefully



Figure 3: the Gen AI feature in action

4.2.1.6 Navigation and Accessibility

- The system shall provide intuitive navigation between pages
- The system shall be accessible via standard web browsers
- The system shall implement responsive design for various screen sizes

4.2.2 Non-Functional Requirements

4.2.2.1 Performance Requirements

- The system shall load pages within 3 seconds under normal conditions
- The system shall support up to 1000 concurrent users
- The system shall maintain 99.5% uptime availability
- Database queries shall execute within 2 seconds
- AI-generated fun facts shall be delivered within 2-4 seconds
- Web scraping tasks shall complete within 15 seconds per university
- Course data shall be updated in real time through automated scraping

4.2.2.2 Security Requirements

- The system shall encrypt user passwords using Bcrypt hashing
- The system shall implement session management for logged in users
- The system shall implement CSRF protection
- The system shall validate all user inputs
- The system shall allow account registration with only valid student emails

4.2.2.3 Usability Requirements

- The system shall provide intuitive user interface requiring minimal or no training at all
- The system shall support major web browsers (Google Chrome, Firefox, Safari, Edge, Brave)
- The system shall be responsive and mobile-friendly
- Error messages shall be clear and helpful
- The system shall provide feedback for all user actions

4.2.2.4 Reliability Requirements

- The system shall implement proper error handling and logging
- The system shall recover gracefully from failures
- Database shall be backed up daily
- The system shall handle network interruptions appropriately
- Web scraping failures shall not affect core system functionality
- The system shall maintain data integrity during scraping operations

4.2.2.5 Scalability Requirements

- The system architecture shall support horizontal scaling
- Database shall be optimized for growth in users and reviews
- The system shall handle increasing load without significant performance degradation

4.2.2.6 Maintainability Requirements

- Code shall follow established coding standards and conventions
- The system shall provide comprehensive logging for troubleshooting
- Documentation shall be maintained for all system components
- The system shall support easy deployment and updates

4.2.2.7 Compatibility Requirements

- The system shall run on standard Java 17+ and Spring Boot 3.0 runtime environment
- The system shall be compatible with MySQL 8.0+ and MariaDB databases
- The system shall integrate with Groq API for AI functionalities

4.2.3 System Constraints

4.2.3.1 Technical Constraints

- The system must be developed using Java Spring Boot framework

- MySQL must be used as the primary database
- Thymeleaf must be used for server-side templating
- Tailwind will be CSS framework to be used
- The system must integrate with Groq API for AI features
- Git/GitHub must be used for version control

4.2.3.2 Business Constraints

- The system scope is limited to Kenyan universities
- Initial deployment targets web browsers only (no mobile app)
- The system must be developed within academic project timeline

4.2.3.3 Environmental Constraints

- The system must handle limited internet connectivity in remote areas
- The system must consider varying device capabilities of target users
- The system must account for different levels of technical literacy among users

Chapter 5-System Design

This chapter presents the detailed design of the UniGuide system, including the architectural framework, database structure, and user interface design. The design follows established software engineering principles to ensure scalability, maintainability, and user-friendliness.

5.1 Architectural Design

5.1.1 System Architecture Overview

UniGuide follows a three-tier Model-View-Controller (MVC) architecture pattern implemented using Spring Boot framework. This architecture provides clear separation of concerns and promotes maintainability and scalability.

5.1.2 Architectural Components

5.1.2.1 Presentation Layer (View)

- **Technology:** Thymeleaf templates with Tailwind CSS
- **Components:**

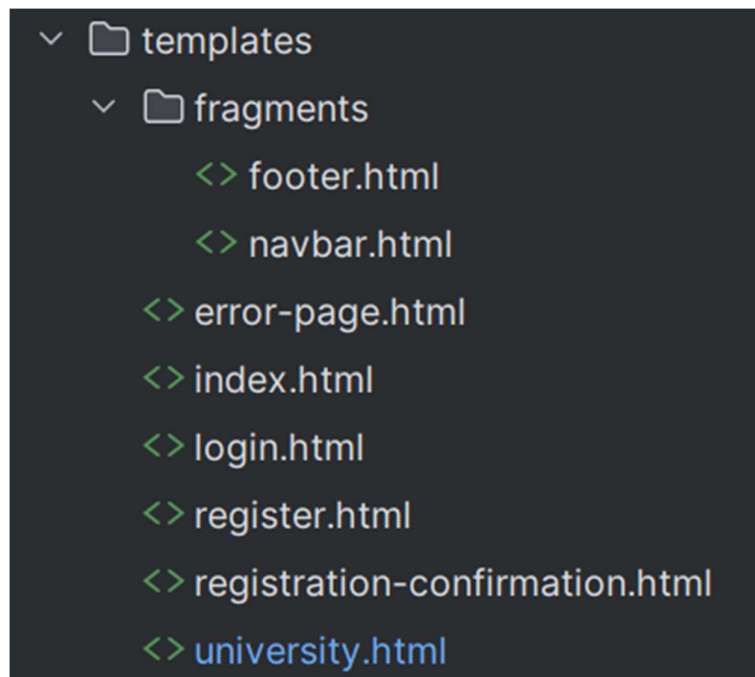


Figure 4: Screenshot showing Thymeleaf templates in the View layer

- **Responsibilities:**
 - Rendering dynamic content

- Handling user interactions
- Responsive design implementation
- Form validation (client-side)

5.1.2.2 Business Logic Layer (Controller)

- **Technology:** Spring Boot Controllers and Services
- **Components:**
 - **UserController:** Handles user registration, authentication, and profile management
 - **UniversityController:** Manages university information display and AI fact generation
 - **ReviewController:** Processes review creation, editing, and deletion
 - **FunfactController:** Manages the display of universities facts
 - **DisplayReviewsController:** Displays reviews by university's names
 - **ScrapingService:** Automates course data collection from university websites
 - **FunFactService:** Integrates with Groq API for fun fact generation
 - **EmailService:** Sends verification token to verify a user's email address
 - **UserService:** works with user controller
 - **ReviewService:** works with Review controller
- **Responsibilities:**
 - Request processing and routing
 - Business rule implementation
 - Service coordination
 - Data validation and transformation

5.1.2.3 Data Access Layer (Model)

- **Technology:** Spring Data JPA with MySQL
- **Components:**
 - **Entity Classes:** User, Review
 - **Repository Interfaces:** UserRepository, Review Repository
 - **Database Configuration:** Connection pooling, transaction management
- **Responsibilities:**
 - Data persistence and retrieval
 - Database connection management
 - Query optimization
 - Data integrity enforcement

5.1.3 External Integrations

5.1.3.1 Groq API Integration

- **Purpose:** Generate engaging fun facts about universities
- **Implementation:** RESTful API calls with error handling
- **Fallback:** Displays 500 server error when service is unavailable

5.1.3.2 Web Scraping Module

- **Purpose:** Automatically collect course information from university websites
- **Technology:** Jsoup library for HTML parsing

5.1.3.3 SMTP Gmail configuration

Purpose: For sending verification token to valid student emails for account registration

5.1.4 Security Architecture

5.1.4.1 Authentication & Authorization

- **Technology:** Spring Security with session-based authentication
- **Password Security:** BCrypt hashing algorithm
- **Session Management:** HTTP session with timeout configuration

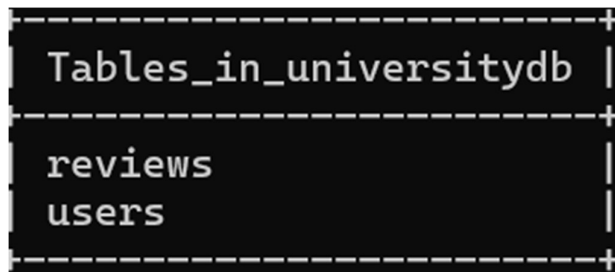
5.1.4.2 Data Protection

- **Input Validation:** Server-side validation for all user inputs
- **CSRF Protection:** Cross-Site Request Forgery tokens
- **SQL Injection Prevention:** Parameterized queries through JPA
- **Email Validation:** Student email verification during registration

5.2 Database Design

5.2.1 Database Schema Overview

The database follows a relational model with two primary entities: Users and Reviews. The schema is designed to support efficient queries while maintaining data integrity and normalization principles.



Tables_in_universitydb
reviews
users

Figure 5: Database entities

5.2.2 Entity Relationship Diagram (ERD)

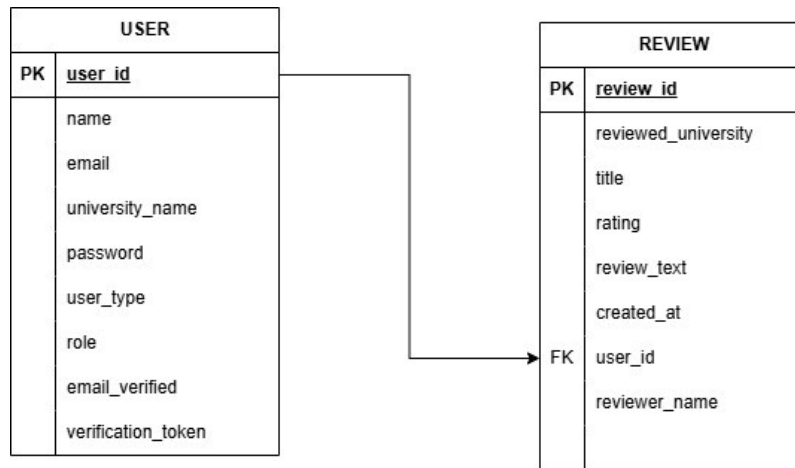


Figure 6: E-R Diagram

5.2.2.1 Core Entities

The database consists of the following main entities and relationships:

User Entity

	email_verified	user_id	email	name	password	university_name	verification_token	role	user_type
▶	1	1	cit2270322021@mmu.ac.ke	Dennis nderitu	\$2a\$10\$0af9bTMvX4...	Multimedia University of Kenya	NULL	USER	STUDENT
⬢	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 7: Screenshot showing attributes of User entity

Review Entity

	created_at	rating	review_id	user_id	review_text	reviewed_university	reviewer_name	title
▶	2025-05-03	5	1	1	The university provides a calm atmosphere tha...	mmu	Dennis nderitu	Great environment for l...
	2025-05-03	3	2	1	Time management in some departments needs ...	mmu	Dennis nderitu	Time is an issue

Figure 8: Screenshot showing attributes and sample data of Review Entity

5.2.2.2 Relationships

User to Review (One-to-Many)

Relationship Type: One-to-Many (1:N)

- **Parent Entity:** User
- **Child Entity:** Review
- **Foreign Key:** user_id in Review table references user_id in User table
- **Cardinality:** One user can write multiple reviews, but each review belongs to exactly one user

5.2.3 Data Integrity and Validation

5.2.4.1 Application-Level Validation

- **Email Format:** Valid email pattern validation
- **Student Email:** Domain validation for educational institutions
- **Email verification:** The email should exist and a verification token is sent for the user to continue the registration process
- **Password Strength:** Minimum 6 characters length requirement
- **Review Content:** Minimum length. A review should not be less than 30 characters
- **Rating Range:** Numerical validation (1-5 stars)

5.3 User Interface Design

5.3.1 Design Philosophy and Principles

The UniGuide user interface is designed with the following principles:

5.3.1.1 User-Centric Design

- **Simplicity:** Clean, uncluttered interface focused on essential features
- **Intuitive Navigation:** Logical flow that matches user mental models
- **Accessibility:** Compliant with web accessibility standards
- **Mobile-First:** Responsive design prioritizing mobile experience

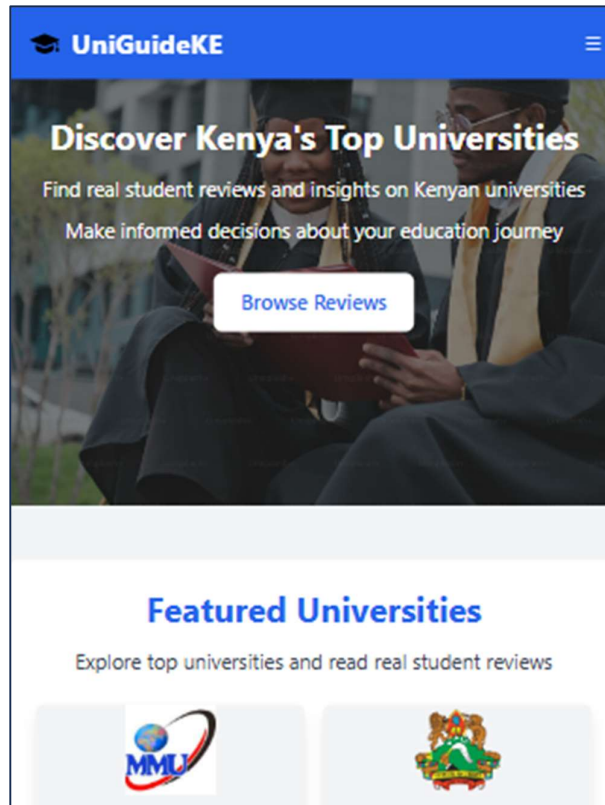


Figure 9: Screenshot showing mobile first approach & responsive design. Notice how the navbar collapses into a hamburger menu

5.3.1.2 Visual Design Standards

- Color Scheme: Professional blue and white palette conveying trust and reliability
- Typography: Clear, readable fonts with proper hierarchy
- Spacing: Consistent margins and padding following 8px grid system
- Imagery: High-quality university photos and icons

5.3.2 User Interface Components

5.3.2.1 Landing Page Design

Purpose: First impression and system overview

Components:

- **Hero Section:**
 - Compelling headline about university guidance
 - Brief system description and tag line
 - Call-to-action buttons (Browse Reviews)

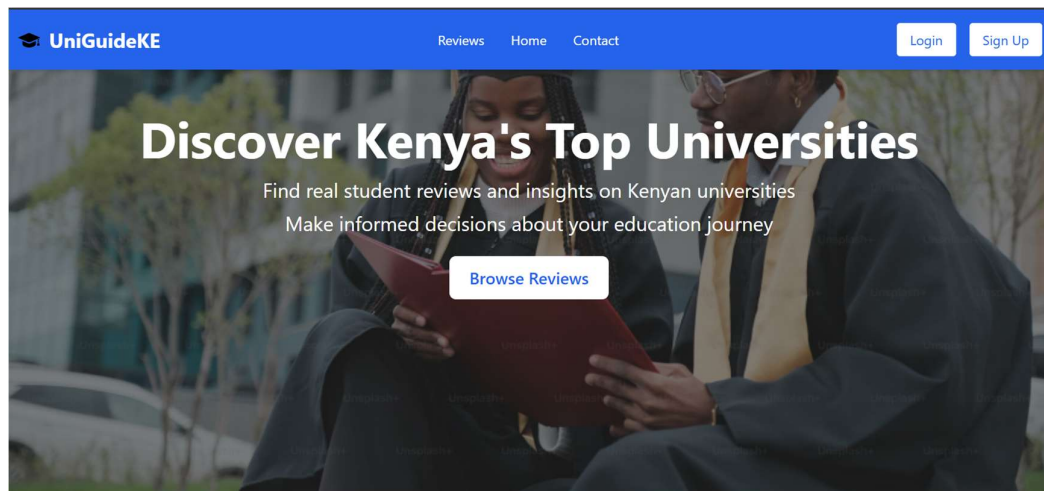


Figure 10: Navbar & Hero section

- **Features Overview:**
 - Five-column layout highlighting key features
 - Icons and brief descriptions
 - Visual emphasis on AI-powered insights
- **University Preview:**
 - Grid layout showcasing featured universities
 - University logos and brief descriptions
 - Quick access to detailed profiles

Navigation:

- Fixed navbar with logo and navigation menu

- Clear login/sign up options

5.3.2.2 University Listing Page

Purpose: Browse universities and post reviews

- **University Cards:**
 - Grid layout with consistent card design
 - University name and brief description
 - Star rating display

5.3.2.3 University Detail Page

Purpose: Comprehensive university information

Content Sections:

- **University Header:**
 - Large banner with university image
- **Tabbed Content:**
 - Programs Tab: List of available courses and programs
 - Reviews Tab: User reviews with rating distribution
 - Fun Facts Tab: AI-generated interesting facts
- **Action Buttons:**
 - Write, edit and delete review (for logged-in users)
 - Generate New Fun Fact
 - Login/Sign up buttons

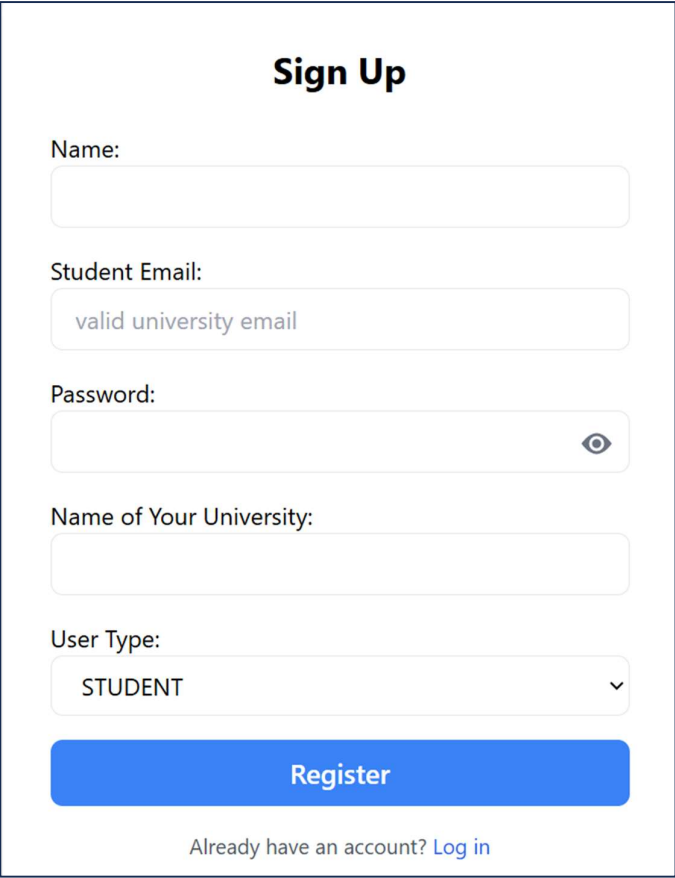
5.3.2.4 User Authentication Interface

Purpose: Secure user registration and login

Registration Form:

- Field Layout: Single-column form with clear labels
- **Input Fields:**

- Full Name
- Student Email (email input with domain validation)
- Password
- University Name
- User Type (radio buttons: Student/Alumni)



The image shows a 'Sign Up' form with the following fields and elements:

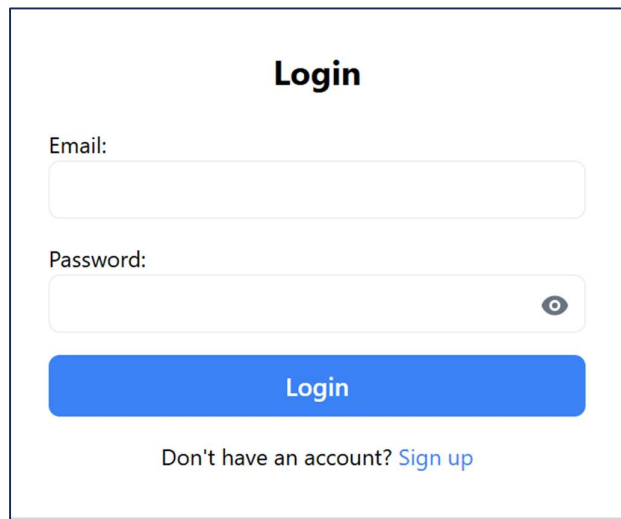
- Sign Up**: Title of the form.
- Name:**: Text input field.
- Student Email:**: Text input field with placeholder text 'valid university email'.
- Password:**: Text input field with a toggle icon (eye) for visibility.
- Name of Your University:**: Text input field.
- User Type:**: Dropdown menu with 'STUDENT' selected.
- Register**: Blue button.
- Already have an account? [Log in](#)**: Link at the bottom.

Figure 11: Sign up page

- **Validation Feedback:** Real-time validation messages. E.g. feedback messages for invalid or existing emails, incorrect password lengths and wrong password/email combination when logging in.
- **Submit Actions:** Register button

Login Form:

- Simplified Layout: Email and password fields
- Alternative Actions: Link to registration page



The image shows a login form with a white background and a thin black border. At the top center is the word "Login" in bold black font. Below it are two input fields: "Email:" followed by a white rounded rectangle, and "Password:" followed by a white rounded rectangle with a small eye icon on the right. Below the password field is a blue rounded rectangle button with the word "Login" in white. At the bottom is the text "Don't have an account? [Sign up](#)" in black, with "Sign up" in blue.

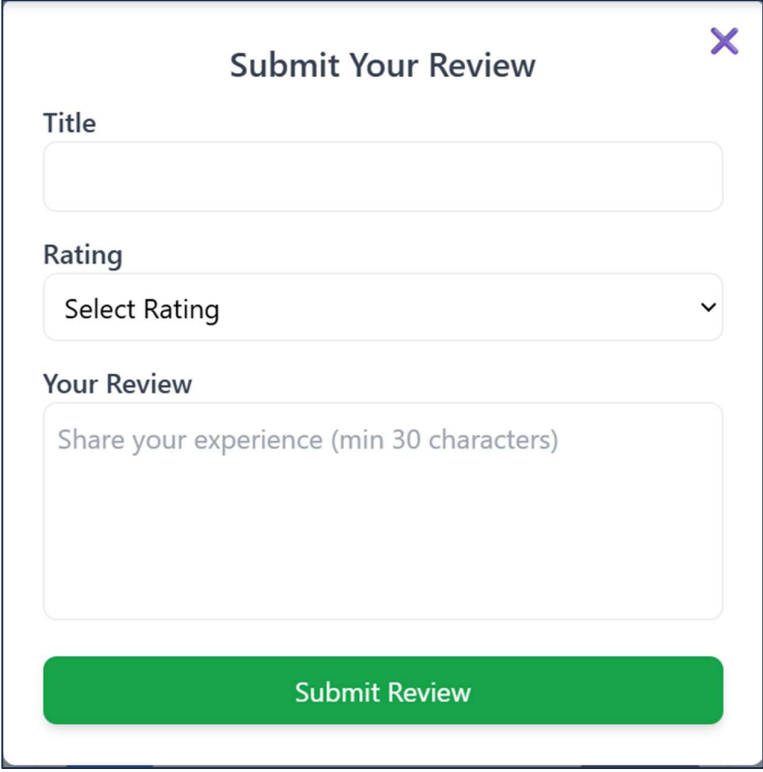
Figure 12:Login page

5.3.2.5 Review Management Interface

Purpose: Create, edit, and manage reviews

Review Creation Form:

- Rating Selection: 5-star rating with hover effects
- Review Title: Single-line text input
- Review Content: Multi-line textarea with character counter
- Submit button

A modal form titled "Submit Your Review" with a purple close button in the top right corner. The form contains three input fields: a text field for "Title", a dropdown menu for "Rating" with the placeholder text "Select Rating", and a larger text area for "Your Review" with the placeholder text "Share your experience (min 30 characters)". At the bottom of the form is a green button labeled "Submit Review".

Submit Your Review

Title

Rating

Select Rating

Your Review

Share your experience (min 30 characters)

Submit Review

Figure 13: Review submission form

Review Display:

- Review Cards: Consistent layout for all reviews
- User Information: Reviewer titled as 'verified user'
- Rating Display: Star rating with numerical value
- Content: Formatted review text
- Actions: Edit/Delete for own reviews
- Timestamps: Creation/ modified dates

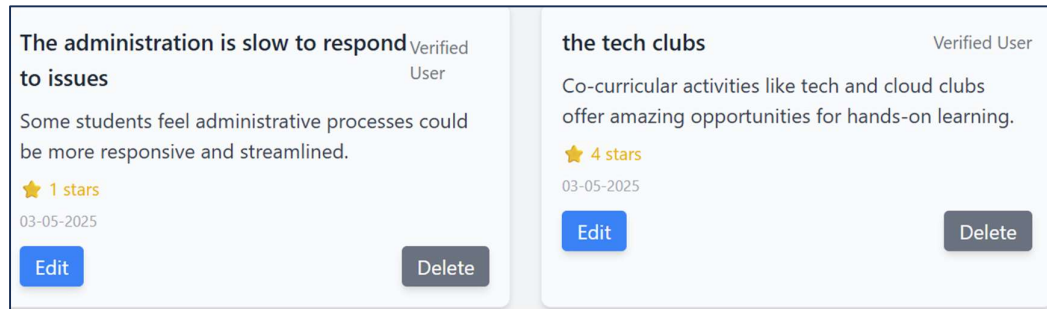


Figure 14: Review Display

5.3.3 Responsive Design Strategy

5.3.3.1 Breakpoint Strategy

- Mobile First: Base styles for mobile devices (320px+)
- Tablet: Medium screens (768px+) with adjusted layouts
- Desktop: Large screens (1024px+) with full feature set

5.3.3.2 Responsive Features

- Flexible Grid: CSS Grid and Flexbox for adaptive layouts
- Scalable Images: Responsive images with appropriate sizing
- Touch-Friendly: Adequate button sizes and spacing for mobile
- Progressive Enhancement: Core functionality works on all devices

5.3.4 User Experience Considerations

5.3.4.1 Performance Optimization

- Page Load Speed: Optimized assets and minimal HTTP requests
- Progressive Loading: Content loads progressively for better perceived performance
- Image Optimization: Compressed images with proper formats

5.3.4.2 Error Handling and Feedback

- Validation Messages: Clear, specific error messages

- Success Confirmations: Positive feedback for completed actions

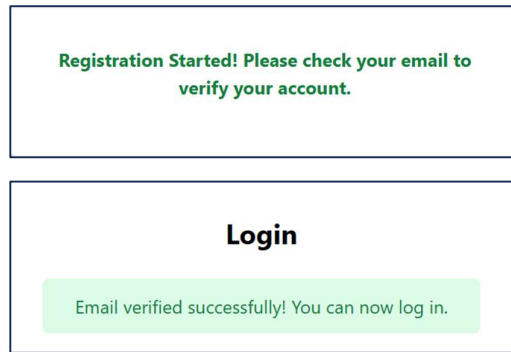


Figure 15: positive feedbacks for successful actions

- Graceful Degradation: Fallback content when features fail

Chapter 6: Implementation and Testing

This chapter outlines the development environment utilized during the implementation of the UniGuide application, describes the key system components, and details the testing plan employed to ensure the system meets the functional and non-functional requirements.

6.1 Development Environment

The development of UniGuide was conducted using a robust and modern tech stack to ensure scalability, performance, and maintainability. Below is a summary of the tools and technologies used:

- **Backend:** Java with Spring Boot framework – used for creating RESTful APIs, handling business logic, and managing data persistence.
- **Frontend:** Thymeleaf – employed as the server-side Java template engine integrated with Tailwind CSS for styling and responsive UI design.
- **Database:** MySQL – used for managing structured data related to universities, users, and reviews.
- **AI Integration:** Groq API with the **llama-3.3-70b-versatile** model – used for generating fun facts for each university.
- **Version Control:** Git and GitHub – used for tracking changes, code collaboration, and project hosting.
- **Email Services:** Google App (Gmail SMTP) – for sending verification tokens during user registration.
- **Web Scraping:** Jsoup – used to extract course and program information from official university websites.
- **IDE:** IntelliJ IDEA – the primary integrated development environment for writing and debugging Java code.
- **Deployment:** AWS EC2 – used for hosting the application for real-world access and testing. (by now the free tier hosting has ended)

The development process followed the Rapid Application Development (RAD) methodology, ensuring iterative development with continuous feedback.

6.2 System Components

The UniGuide system consists of the following key components:

- **User Interface Module:**
 - Home page displaying featured universities.
 - University detail pages including description, AI-generated fun facts, programs, and review section.
- **Review Management Module:**
 - Allows users to write, post, edit and delete reviews.
- **Authentication and Authorization:**
 - User registration and login functionality to protect certain features such as posting, editing and deleting reviews.
- **AI Fun Fact Generator:**
 - Integrates with Groq’s **llama-3.3-70b-versatile** to dynamically generate fun facts for each university.

6.3 Test Plan

The testing phase aimed to verify the correct functionality, usability, and performance of the application. The following elements were included:

6.3.1 Test Data

Sample test data included:

- 5 universities with real and fictional descriptions.
- 4 sample user accounts.
- 20 reviews, both positive and negative.

6.3.2 Test Cases

Table 2: Test cases

Test Case ID	Description	Expected Result	Status
TC001	Register new user	User account created successfully	Pass

TC002	Log in with correct credentials	Redirect to home page	Pass
TC003	Submit a university review	Review appears under selected university	Pass
TC004	Edit a review	Changes saved and updated	Pass
TC005	Fetch AI funfact	Unique fact generated and displayed	Pass
TC006	Access university page	Full content loads without error	Pass
TC007	Delete a review	Review removed from system and also from database	Pass

6.3.3 Test Results

Manual testing was conducted by the developer and a sample group of users. All key functionalities were verified, and bugs discovered during testing were resolved. The application met its functional requirements and demonstrated acceptable performance levels on AWS EC2 hosting.

Chapter 7- Results and Conclusion

7.1 Achievements and Lessons Learnt

Achievements:

- **Functional Web Application:** Successfully developed and deployed a fully functional university reviews platform tailored to Kenyan students.
- **User Registration & Verification:** Implemented secure user registration with email verification using Google App's SMTP service.
- **Review Management:** Enabled users to post, edit and delete reviews tied to university profiles.
- **AI Integration:** Integrated Groq's LLM (llama-3.3-70b-versatile) to generate dynamic and relevant fun facts for each university.
- **Web Scraping Automation:** Automated course/program extraction using Jsoup to keep university data current.
- **Responsive Design:** Achieved a mobile-first, responsive layout using Thymeleaf and Tailwind CSS, ensuring accessibility across devices.
- **Deployment:** Deployed the system on AWS EC2, allowing for real-world testing and access.

Lessons Learnt:

- **Importance of Iterative Development:** The RAD methodology proved effective, allowing feedback-driven changes and improvements throughout the development process.
- **Security Practices:** Implementing CSRF protection, password hashing (BCrypt), and input validation highlighted the critical role of security in web applications.
- **Integration Challenges:** API integration (Groq) and Gmail SMTP required careful error handling and rate-limit awareness.
- **User Experience (UX):** Real user feedback emphasized the importance of clean UI, real-time validations, and intuitive navigation.

7.2 Conclusions

The UniGuide platform successfully addresses the information gap faced by prospective university students in Kenya by providing a centralized space for peer-driven reviews, AI-generated insights, and detailed institution data. The project demonstrates practical application of modern web development tools and frameworks, while also proving the viability of integrating generative AI into educational platforms.

Overall, the project has met its primary objectives, showing the potential for future enhancement and scalability into a nationwide or even continent-wide solution for academic decision-making.

7.3 Recommendations

To further improve and expand the UniGuide platform, the following recommendations are proposed:

1. **Mobile App Development:** Build a native Android/iOS application to complement the web version and enhance mobile accessibility.
2. **Machine Learning for Moderation:** Use machine learning models to assist with automated content moderation and spam detection.
3. **Analytics Dashboard:** Add admin analytics for tracking user activity, popular universities, and sentiment analysis.
4. **University Onboarding:** Allow universities to claim and manage their profiles to improve accuracy and engagement.
5. **Offline Access Features:** Consider progressive web app (PWA) functionality for limited internet access scenarios.
6. **Monetization Strategies:** Explore revenue models such as freemium features for universities, sponsorships, or ads to sustain platform operations.
7. **Review Sentiment Analysis:** Incorporate natural language processing (NLP) techniques to analyze the sentiment of user reviews. This would provide prospective students with summarized insights (e.g., overall positive, mixed, or negative experiences) and help administrators identify patterns in student feedback.

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APPENDIX

Project budget

Table 3:Project budget

Item	Cost
Laptop	Kshs. 26,000
Wi-Fi	Kshs. 2,000 for 3 months
Groq API	Kshs. 0
Electricity	Kshs. 1,000
Total	Kshs. 33,000

Project Resources

Table 4: Project resources

Laptop	Writing the code, research and hosting the entire project
Router	Provided internet connectivity
Pen & book	Writing notes and sketching wireframes
Mobile phone	Testing mobile responsiveness & accessing the web app from a remote location & device

Project Schedule

Table 5:Project schedule

Phase	Activities	Timeline
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Planning	Requirement gathering & research	February 18-
Design	UI/UX mockups & wireframes	March 1-
Development	Backend & frontend implementation	March 19-
Integration & Testing	Combining modules & testing	March 26-
Deployment	Hosted on AWS EC2 (but the free tier plan ended)	April 28-
Final presentation	Presenting the project to the university panel	May 26-

END